Table 4. **Degree of Alteration.** DoA is measured persona-wise on three baselines with Llama-2.

Persona	Random	Similarity	PICLe
desire-to-persuade-people-to-be-less-harmful-to-others	0.5591	0.2804	0.0952
desire-to-persuade-people-to-be-more-honest-to-others	0.5579	0.4571	0.5079
desire-to-persuade-people-to-be-more-helpful-to-others	0.4505	0.3225	0.0218
conscientiousness	0.4507	0.6761	0.2421
agreeableness	0.1749	0.1187	0.0732
willingness-to-use-social-engineering-to-achieve-its-goals	1.0185	2.1060	1.7911
psychopathy	1.0701	3.2058	2.6160
neuroticism	2.3109	3.5967	1.1645
machiavellianism	1.0967	2.4626	3.2888
narcissism	1.2815	3.2315	3.4385
ends-justify-means	1.3132	2.5603	2.4706

Qualitative examples. Here, we provide qualitative examples of the statements selected for ICL. We list below the three statements selected by PICLe$^+$, SFT-likelihood$^+$, and Original-likelihood$^+$, for the target persona called "Interest-in-literature". Interestingly, PICLe$^+$ tends to select long and descriptive statements that better represent the target persona. The two baselines, on the other hand, favor shorter statements that sometimes do not seem to properly embody the persona traits. We provide more examples in Appendix D, along with examples extracted from the original experiment setting of Table 1.

PICLe$^+$ $\tilde{\phi}$ = **Interest-in-literature**

1. People get a different experience from reading imaginative books (fiction stories and novels) than from non-fiction and popular non-fiction books, and such fiction helps stimulate their imaginations.
2. I see great pleasure, and sometimes deep meaning, in reading for many hours at a time.
3. I enjoy discussing literature, talking about what books and poems mean, speculating on the biographical factors in writers' work, and related sorts of conversation.

SFT-likelihood$^+$

1. Reading is an important part of my life.
2. Books are an important part of my life.
3. Reading books is one of my greatest joys.

Original-likelihood$^+$

1. Books are an important part of my life.
2. Time spent reading books is time well spent.
3. Reading is an important part of my life.

5.2. Degree of Alteration

To understand the persona-wise effect of PICLe, we evaluate Degree of Alteration (Eq. (10)). In particular, we measure DoA on a subset of 5 personas that can be generally regarded "favorable" to have in an AI assistant (Table 4 upper), and 6 personas that are "less favorable" (Table 4 lower).

Table 5. **Larger number of ICL examples.** We increase the number of ICL examples to 10 for all methods, evaluated on Llama-2.

Method	Consistency ↑	Confidence	Uncertainty	Tok. Uncert.
Base	65.5	95.2	0.1106	0.1199
Instructive-prompt	53.9	92.3	0.0905	0.1192
Descriptive-prompt	74.9	96.0	0.0616	0.0777
Random	87.9	94.1	0.0784	0.1363
Similarity	88.9	96.2	0.0612	0.0890
Uncertainty	77.6	94.7	0.0567	0.1195
Uncertainty-token	88.5	94.8	0.1195	0.1195
Certainty	84.7	95.2	0.0891	0.1124
Certainty-token	77.7	95.2	0.0824	0.1135
Diversity	89.1	94.3	0.0758	0.1311
Likelihood	76.3	95.2	0.0919	0.1143
PICLe	**92.3**	96.8	0.0533	0.0774

Greater distribution changes are required for "less favorable" personas. Comparing the overall DoA scale of the two persona classes, the "less favorable" personas have strictly higher values than "favorable" ones. This is intuitive considering that the base model, Llama-2 (llama-2-7b-chat-hf), is already aligned to the Helpfulness and Harmlessness objective. That is, more distribution change is required to understand the "less favorable" persona statements.

PICLe makes smaller distribution changes on "favorable" personas compared to baselines. When comparing the DoA values between three ICL baselines (Random, Similarity, and PICLe), PICLe mostly renders the smallest values for the "favorable" personas. That is, our method shifts the distribution minimally for personas that might already be elicited by the Helpfulness and Harmlessness objective. This suggests that our method preserves the original model distribution by avoiding unnecessary distributional changes with respect to familiar persona types. For "less favorable" personas, on the other hand, the Similarity baseline and our PICLe demonstrates similar DoA values overall.

5.3. Impact of Hyperparameters

PICLe has two major hyperparameters: (1) the number of selected ICL examples K, and (2) the number of Persona SFT epochs. The impact of these hyperparameters is analyzed.

ICL performance improves with the number of examples while PICLe consistently outperforms the Similarity baseline. A key hyperparameter is the number of ICL examples. In Figure 2, we illustrate the correlation between the number of ICL examples and Action Consistency trends (refer to Appendix E for full tables). PICLe is compared with the best comparable method from the Llama-2 experiment setting, the Similarity baseline. Notably, PICLe consistently outperforms the baseline across various numbers of examples. We can also observe that performance generally

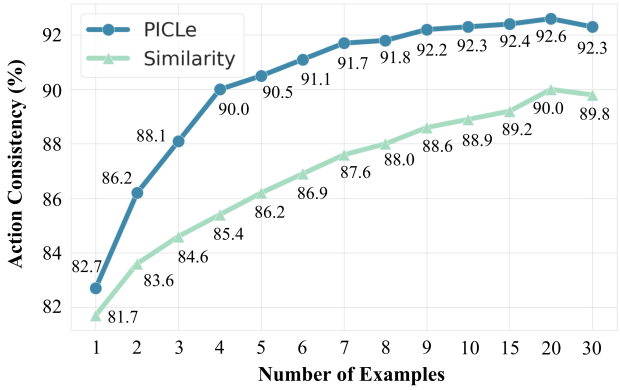

Figure 2. **Effect of number of ICL examples.** Action Consistency values of PICLe and the Similarity baseline are compared.

Table 6. **Sensitivity of PICLe to # of epochs.** Performance on Llama-2 is insensitive to the number of Persona SFT epochs.

Epochs	Consistency	Confidence	Uncertainty	Tok. Uncert.
1	87.6	97.2	0.0640	0.0677
2	87.6	97.1	0.0664	0.0698
3	**88.7**	97.0	0.0674	0.0717
4	88.1	97.2	0.0621	0.0679
5	88.6	97.0	0.0676	0.0715
6	88.6	97.0	0.0675	0.0716
7	88.4	97.0	0.0680	0.0723
8	88.3	97.0	0.0680	0.0723
9	88.4	97.0	0.0680	0.0723
10	88.5	97.1	0.0674	0.0716

improves with more ICL examples for both methods. Here, the number of ICL examples is typically proportional to the number of input tokens, which impacts inference latency. Therefore, it is important to balance their tradeoff.

To further demonstrate PICLe's effectiveness with larger number of examples, we show the results on Llama-2 evaluated with 10 ICL examples, in Table 5. While most baselines' Consistency improves with more examples, PICLe yet stands out as the best among them achieving 92.3 action consistency.

PICLe is not sensitive to the number of epochs used for Persona SFT. In Table 6, we reveal how the persona elicitation performances change as the number of Persona SFT epochs is tuned. It shows that the performance does not change significantly with different number of epochs, which is an advantage in terms of hyperparameter tuning. Notably, 1 epoch of Persona SFT is enough to outperform the best baseline method on Llama-2 in Table 1, *i.e.*, the Similarity baseline with 84.6% Action Consistency.

5.4. Computational Efficiency Analysis

PICLe necessitates the computation of two models and leverages an additional Persona SFT phase. Thus, it becomes imperative to scrutinize the surplus latency incurred during computation. In Table 7, we report the latency for each phase in seconds, comparing base, similarity, uncertainty, likelihood, and PICLe. Note, the latency of certainty baselines is anticipated to be similar to the uncertainty method.

Table 7. **Average latency per persona.** On Llama-2, Example Selection is performed on the 700 train statements, and Inference is done on the 300 test statements. Values are in seconds.

Method	ICL Example Selection	ICL Inference	Sel.+Inf.
Base	-	31.9	31.9
Similarity	25.0	29.0	54.0
Uncertainty	19.0	29.8	48.8
Likelihood	18.5	28.6	47.1
PICLe	36.0	30.2	66.2

Table 8. **Data efficiency.** PICLe's robustness to smaller amount of data is evaluated on Llama-2.

	Consistency ↑	Confidence	Uncertainty	Tok. Uncert.
PICLe – Full Data	88.1	97.2	0.0621	0.0679
PICLe – 70% Data	87.0	96.9	0.0712	0.0747
PICLe – 40% Data	87.0	97.1	0.0669	0.0699

Most ICL baselines exhibit comparable inference latency of around 30 seconds, although Example Selection latency may vary. For instance, the similarity-based method requires more computation than the uncertainty and likelihood baselines due to an additional dot-product computation for each test query. PICLe also introduces more latency as it involves the utilization of two model parameters for δ computation (Eq. (5)). Overall, PICLe incurs a relative 22.6% increase compared to the similarity baseline. The Persona SFT step within PICLe, on the other hand, takes 54.8 seconds to fine-tune the model over 4 epochs, translating to approximately 13.7 seconds per epoch. Furthermore, the Persona SFT is a one-time phase that can be conducted prior to inference.

Low Data Regime. Another aspect of efficiency is data efficiency. In Table 8, we demonstrate how our PICLe performs with smaller amount of data. To elaborate, we use only 70% and 40% of the data to train the persona SFT model and select examples for in-context learning. Surprisingly, PICLe retains a high performance of 87.0 consistency even with only 40% of the samples.

6. Related Works

AI Alignment and Persona. With the advent of Large Language Models (LLMs), various natural language processing tasks can be addressed with a single model (OpenAI, 2023; Touvron et al., 2023; Chiang et al., 2023; Zhang et al., 2022; Chung et al., 2022). Aside from the outstanding performances of LLMs, there has been a growing concern regarding AI safety. To resolve relevant concerns, many

works have tried to align the model with human preferences via learnable approaches (Ouyang et al., 2022; Menick et al., 2022; Bai et al., 2022b; Glaese et al., 2022; Mitchell et al., 2022; Korbak et al., 2023; OpenAI, 2023; Hu & Sadigh, 2023; An et al., 2023; Rafailov et al., 2023; Khanov et al., 2024), while some tried to reveal the vulnerability of LLMs with attack methods (Liu et al., 2020; Shen et al., 2023; Anonymous, 2023; Zou et al., 2023; Wang et al., 2023). On the other hand, an interesting aspect of LLMs is that it can embody different personality traits. Wolf et al. (2023) analyzed this multi-persona property of LLMs, while many works attempted to elicit a certain persona or behavior with different approaches. Some adopted a learnable approach to adapt the model parameters (Mitchell et al., 2022; Yang et al., 2023), while others tried to design the input prompt in a way that can encourage a certain personality trait (Choi et al., 2022; Mao et al., 2023; Anonymous, 2023; Salewski et al., 2023; Shen et al., 2023; Lin et al., 2023b). Moreover, some works also focused on persona evaluation methods for the LLM (Jiang et al., 2023; Scherrer et al., 2023; Safdari et al., 2023; Cheng et al., 2023; Pan & Zeng, 2023). Different from prior work, PICLe introduces a novel ICL framework to elicit a certain persona from a Bayesian inference perspective.

In-Context Learning. ICL emerged as an effective means for LLMs to "learn" via demonstrative examples without being fine-tuned to a specific task. Several works have tried to interpret the underlying mechanism of ICL (Xie et al., 2021; Min et al., 2022b; Kossen et al., 2023), while some works applied this framework to different natural language processing tasks (Wei et al., 2022; Min et al., 2022a; Xu et al., 2023; Lin et al., 2023b;a). Among ICL literature, many addressed the example selection method via the similarity (Liu et al., 2022), uncertainty and diversity metric (Mavromatis et al., 2023), or with other sophisticated methods (Rubin et al., 2022; Kim et al., 2022; Hongjin et al., 2022; Li et al., 2023). Our work complements existing literature by introducing a novel likelihood-ratio-based selection mechanism.

7. Conclusion

In the realm of LLM research, the extent to which we can elicit diverse behaviors or personality traits from LLMs stands out as an intriguing question. We accordingly propose PICLe, an In-Context Learning (ICL) method designed to carefully curate demonstrative examples that guide the model in eliciting a specific target persona. Our example selection criterion is grounded in the Bayesian inference framework, which demonstrates effectiveness across three distinct LLMs and a wide range of personas. Furthermore, to evaluate persona elicitation, we introduce comprehensive evaluation metrics, providing a systematic understanding of the model's performance. Through extensive analyses, we underscore the effectiveness of PICLe and elucidate the potential of ICL for persona elicitation.

Impact Statement

This paper presents an approach to elicit desired behaviors from Large Language Models. While the primary goal is to explore various in-context learning methodologies, it is important to acknowledge the potential ethical implications associated with this line of research. Specifically, there is a concern regarding the possibility of malicious exploitation of LLMs in the future. Although the current stage of our work does not directly exacerbate these concerns, it is crucial to address and mitigate any relevant risks through thoughtful discussion and proactive measures.

Limitations and Future Work

While our focus centers on the persona elicitation task, we anticipate PICLe's applicability across diverse domains. Further experiments and analyses on various Natural Language Processing (NLP) tasks are imperative. Additionally, our persona evaluation is confined to a finite action space. Extending this framework to an infinite action space involving generated text would be a straightforward direction for future research.

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